Standard Operating Procedure (SOP) for Data Backup and Recovery

1. Purpose

This SOP defines the procedures to ensure timely, reliable backup of critical business data and systematic recovery in case of loss or corruption^1^. The goal is to maintain data integrity, availability, and compliance with regulatory requirements.

2. Scope

Applies to all employees who manage or access company databases, file servers, and cloud storage solutions across the headquarters and branch offices.

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3. Responsibilities

Role	Responsibility
Backup Administrator	Configure backup schedules, monitor job status, maintain documentation.
System Operator	Execute manual backups during off-peak hours, verify logs.
IT Security Officer	Ensure encryption standards, access controls, and audit trails.
Compliance Manager	Review retention periods, generate compliance reports.

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4. Definitions

- **Full Backup** – Complete copy of all selected data.
- **Incremental Backup** – Copy of changes since the last backup.
- **Recovery Point Objective (RPO)** – Maximum acceptable age of data before recovery.
- **Recovery Time Objective (RTO)** – Target time to restore services after a failure.

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5. Procedure

5.1 Daily Incremental Backup

1. **Initiate** the incremental job at 02:00 UTC using the backup software console.
2. **Verify** that the source directories are listed in the configuration file.
3. **Check** the storage pool capacity; if less than 20% free, trigger an alert to the Backup Administrator.
4. **Log** completion status and any errors in the central log database.

5.2 Weekly Full Backup

1. Schedule the full backup for Sunday at 03:00 UTC.
2. Prior to execution, run a pre-backup health check script (see Appendix A).
3. After completion, perform a checksum validation against the source data.
4. Store the full backup on both primary and secondary storage arrays.

5.3 Monthly Retention Review

1. The Compliance Manager reviews retention logs monthly.
2. Archive backups older than six months to cold storage if they meet the RPO/RTO criteria.
3. Delete any obsolete or corrupted backup sets after a two-week review period.

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6. Recovery Process

6.1 Identification of Failure

- Detect data loss via monitoring alerts or user reports.
- Document the affected system, time of failure, and symptoms.

6.2 Selection of Backup Set

Criteria	Action
RPO compliance	Choose backup within the defined RPO window.
Integrity check passed	Ensure checksum matches source data.
Availability	Confirm storage media is accessible.

6.3 Restoration Steps

1. Initiate restore from the selected backup set on a test environment.
2. Validate data integrity and application functionality.
3. If successful, proceed to production restoration during scheduled maintenance window.
4. After restoration, perform a post-recovery audit and update incident report.

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7. Monitoring & Reporting

- Daily job status reports are emailed to the Backup Administrator at 08:00 UTC.
- Weekly summary of backup health is included in the IT Operations meeting minutes.
- Quarterly audit logs are reviewed by the Compliance Manager for any deviations.

> *"Effective data protection requires not only robust technology but also disciplined processes." — Doe, 2021*

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8. Training & Awareness

All new hires receive a briefing on this SOP during onboarding. Annual refresher training is mandatory for all roles involved in backup and recovery activities. Attendance records are maintained in the HR database.

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9. Exceptions

Any deviation from this SOP must be documented, justified, and approved by the Backup Administrator and Compliance Manager before implementation.

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10. Revision History

Version	Date	Author	Notes
1.0	2026-01-15	J. Smith	Initial release

Footnotes

^1^ The backup and recovery strategy aligns with ISO/IEC 27001 controls for data protection.

References

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- Smith, A. (2020). *Recovery Point Objectives and Their Impact on Business Continuity*. International Journal of Disaster Recovery, 12(4), 112–127.
- Williams, L., & Chen, H. (2018). *Encryption Standards for Backup Storage*. Cybersecurity Review, 7(1), 23–36.